

$$1) f_n = 125 \text{ cpm} \\ = \frac{25}{12} \text{ Hz}$$

$$f_d = 120 \text{ cpm} \\ = 2 \text{ Hz}$$

$$\omega_d = \omega_n \sqrt{1 - \xi^2}$$

$$2\pi f_d = 2\pi f_n \sqrt{1 - \xi^2}$$

$$2^2 = \left(\frac{25}{12}\right)^2 (1 - \xi^2)$$

$$\frac{576}{625} = 1 - \xi^2$$

$$\xi = 0.28$$

$$\xi = \frac{c}{c_c} = \frac{c}{2\sqrt{km}} = \frac{c}{2m\omega_n}$$

$$\therefore c_c = 2\sqrt{km} = 2m\omega_n = \frac{2k}{\omega_n}$$

$$\therefore 0.28 = \frac{c}{2(2)(2\pi)\left(\frac{25}{12}\right)}$$

$$c = \frac{14}{3} \pi$$

$$\approx 14.66 \text{ N s m}^{-1}$$

$$2a) \frac{x_1}{x_3} = \frac{x_1}{x_2} \frac{x_2}{x_3} = \left(\frac{x_1}{x_2} \right)^2 = \frac{40}{10}$$

$$\therefore \left(\frac{x_1}{x_2} \right)^2 = \frac{40}{10}$$

$$\frac{x_1}{x_2} = \sqrt{4}$$

$$\frac{x_1}{x_2} = 2$$

$$\delta = \ln\left(\frac{x_1}{x_2}\right) = \ln 2 \approx 0.693$$

b)

$$\zeta = \sqrt{\frac{\delta^2}{4\pi^2 + \delta^2}} = \sqrt{\frac{(\ln 2)^2}{4\pi^2 + (\ln 2)^2}} \approx 0.109652581$$

$$\approx 0.1097$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}} = \frac{2\pi(1)}{\sqrt{1 - 0.1097^2}} \approx 6.321302921$$

$$\approx 6.321 \text{ rad s}^{-1}$$

$$2c) F = \frac{\mu A v}{h}$$

$$= \frac{\mu A}{h} v$$

$$= c \dot{x}, \quad c = \frac{\mu A}{h}$$

$$\xi = \frac{c}{c_c}$$

$$c = \xi c_c = \xi (2m\omega_n)$$

$$\therefore c = 2m\omega_n \xi$$

$$\frac{\mu A}{h} = 2m\omega_n \xi$$

$$\mu = \frac{2m\omega_n \xi h}{A}$$

$$= \frac{2(28 \times 10^{-3})(6.321)(0.1097)(5 \times 10^{-3})}{\pi(10 \times 10^{-3})(0.25)}$$

$$= 0.0247118722$$

$$\approx 24.7 \text{ mPa s}$$

$$3) J_0 \ddot{\theta} = \sum M_0$$

$$\sum M_0 = -k_\theta \theta - c_\theta \dot{\theta}$$

$$\therefore J_0 \ddot{\theta} = -k_\theta \theta - c_\theta \dot{\theta}$$

$$J_0 \ddot{\theta} + c_\theta \dot{\theta} + k_\theta \theta = 0$$

$$J_0 = \frac{1}{3} mL^2$$

$$= \frac{1}{3} (36) (0.75)^2$$

$$= 6.75 \text{ kg m}^2$$

$$\omega_n = \sqrt{\frac{k_\theta}{J_0}}$$

$$= \sqrt{\frac{10}{6.75}}$$

$$= \frac{2\sqrt{30}}{9}$$

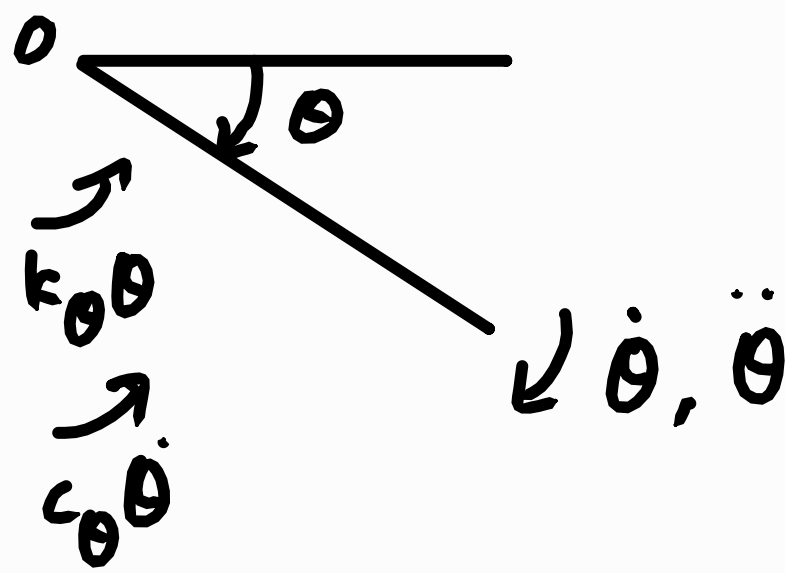
$$\approx 1.217 \text{ rad s}^{-1}$$

$$\zeta = \frac{c_\theta}{c_{\theta_c}} = \frac{c_\theta}{2\sqrt{k_\theta J_0}}$$

$$\therefore 1 = \frac{c_\theta}{2\sqrt{10(6.75)}}$$

$$c_\theta = 3\sqrt{30}$$

$$\approx 16.43 \text{ Nms rad}^{-1}$$



3) For critically damped oscillations:

$$\theta(t) = (C_1 + C_2 t) e^{-\omega_n t} \quad (1)$$

$$\dot{\theta}(t) = C_2 e^{-\omega_n t} + (C_1 + C_2 t) (-\omega_n) e^{-\omega_n t}$$

When $\theta(0) = \frac{\pi}{2}$, $\dot{\theta}(0) = 0$,

$$\theta(0) = C_1$$

$$\therefore C_1 = \frac{\pi}{2}$$

$$\dot{\theta}(0) = C_2 + (C_1 + 0) (-\omega_n)$$

$$0 = C_2 - \omega_n C_1$$

$$C_2 = \omega_n C_1$$

$$= \frac{2\sqrt{30}}{9} \left(\frac{\pi}{2} \right)$$

$$= \frac{\sqrt{30} \pi}{9}$$

$$\approx 1.9119$$

$$\therefore \text{From (1): } \theta(t) = \left(\frac{\pi}{2} + \frac{\sqrt{30} \pi}{9} t \right) e^{-\omega_n t}$$

$$\text{When } \theta(t) = 1^\circ = \frac{\pi}{180}$$

$$\frac{\pi}{180} = \left(\frac{\pi}{2} + \frac{\sqrt{30} \pi}{9} t \right) e^{-\frac{2\sqrt{30}}{9} t}$$

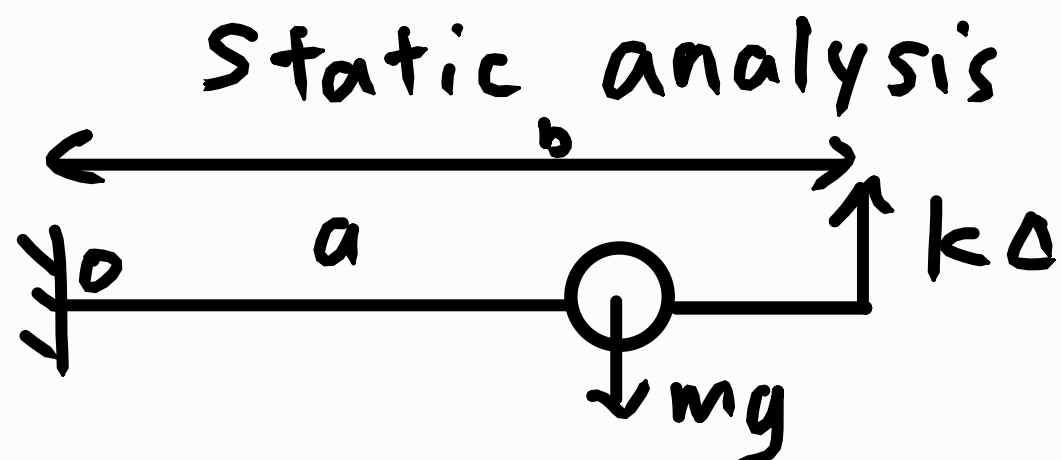
Solving using an iterative solver:

$$t = 5.354239868 \approx 5.355_s$$

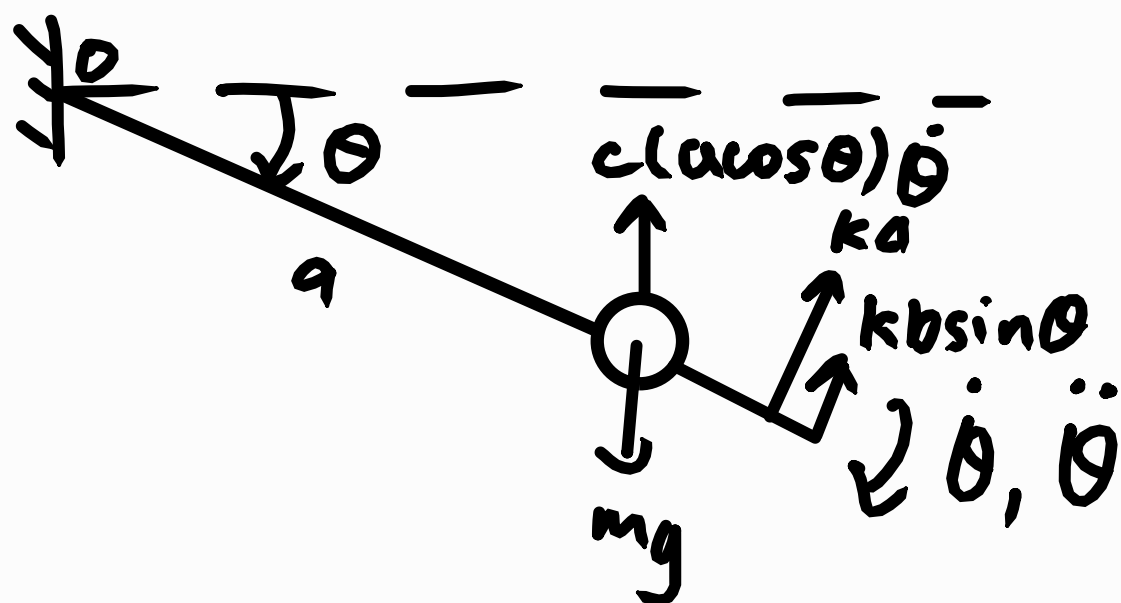
$$4) \sum M_o = 0$$

$$\curvearrowright mga = k\Delta b$$

$$mga - k\Delta b = 0 \quad (1)$$



Dynamic analysis



$$J_o \ddot{\theta} = \sum M_o$$

$$J_o = ma^2$$

$$\begin{aligned} \curvearrowright \sum M_o &= mga \cos \theta - k(\Delta + b \sin \theta) b \cos \theta - c(a \cos \theta) \dot{\theta} a \cos \theta \\ &= mga \cos \theta - kb \cos \theta (\Delta + b \sin \theta) - ca^2 \cos^2 \theta \dot{\theta} \end{aligned}$$

Using small angle approximation,

$$\cos \theta \approx 1, \quad \sin \theta \approx \theta$$

$$\sum M_o = mga - kb(\Delta + b\theta) - ca^2 \dot{\theta}$$

$$= mga - kb\Delta - kb^2\theta - ca^2 \dot{\theta}$$

$$\therefore ma^2 \ddot{\theta} = mga - kb\Delta - kb^2\theta - ca^2 \dot{\theta}$$

$$ma^2 \ddot{\theta} + ca^2 \dot{\theta} + kb^2\theta = mga - kb\Delta$$

$$ma^2 \ddot{\theta} + ca^2 \dot{\theta} + kb^2\theta = 0 \quad \therefore (1)$$

$\therefore$ The equation of motion is:

$$ma^2 \ddot{\theta} + ca^2 \dot{\theta} + kb^2\theta = 0$$

$$4) \quad ma^2 \ddot{\theta} + ca^2 \dot{\theta} + kb^2 \theta = 0$$

$$J_0 \ddot{\theta} + C_\theta \dot{\theta} + K_\theta \theta = 0$$

Natural frequency:

$$\omega_n = \sqrt{\frac{K_\theta}{J_0}} = \sqrt{\frac{kb^2}{ma^2}} = \frac{b}{a} \sqrt{\frac{k}{m}}$$

Critical damping constant:

$$(C_\theta)_c = 2\sqrt{K_\theta J_0} = 2\sqrt{kb^2 ma^2} = 2ab\sqrt{km}$$

Damping ratio:

$$\xi = \frac{C_\theta}{(C_\theta)_c} = \frac{ca^2}{2ab\sqrt{km}} = \frac{ca}{2b\sqrt{km}}$$

Natural frequency of the damped oscillation:

$$\omega_d = \omega_n \sqrt{1 - \xi^2}$$

$$= \frac{b}{a} \sqrt{\frac{k}{m}} \sqrt{1 - \left(\frac{ca}{2b\sqrt{km}}\right)^2}$$

$$= \frac{b}{a} \sqrt{\frac{k}{m} \left[1 - \left(\frac{ca}{2b\sqrt{km}}\right)^2\right]}$$

$$5) \quad m=1, \quad k=9, \quad c=1, \quad x(0)=30 \\ \dot{x}(0)=0$$

$$c_c = 2\sqrt{km} = 2\sqrt{9(1)} = 6$$

$$\xi = \frac{c}{c_c} = \frac{1}{6} < 1 \quad (\text{Under-damped})$$

Response of under-damped system:

$$x(t) = e^{-\xi\omega_n t} X \sin(\omega_d t + \phi)$$

$$\dot{x}(t) = -\xi\omega_n e^{-\xi\omega_n t} X \sin(\omega_d t + \phi)$$

$$+ e^{-\xi\omega_n t} \omega_d X \cos(\omega_d t + \phi)$$

$$= -\xi\omega_n x(t) + e^{-\xi\omega_n t} \omega_d X \cos(\omega_d t + \phi)$$

$$\text{When } x(0) = 30 \times 10^{-3},$$

$$30 = 1(X) \sin(\phi)$$

$$X \sin \phi = 30 \quad (1)$$

$$\text{When } \dot{x}(t) = 0,$$

$$0 = -\xi\omega_n(30) + (1)\omega_d X \cos \phi$$

$$X \cos \phi = \frac{30\xi\omega_n}{\omega_d} = \frac{30\xi\omega_n}{\omega_n \sqrt{1-\xi^2}}$$

$$\therefore X \cos \phi = \frac{30\xi}{\sqrt{1-\xi^2}} \quad (2)$$

5) Dividing (1) by (2)

$$\frac{X \sin \phi}{X \cos \phi} = \frac{\cancel{30}}{\cancel{30\xi} \over \sqrt{1-\xi^2}}$$

$$\tan \phi = \frac{\sqrt{1-\xi^2}}{\xi}$$

$$\phi = \arctan \left(\frac{\sqrt{1-(\frac{1}{6})^2}}{\frac{1}{6}} \right)$$

$$= 80.40593177^\circ$$

$$\approx 80.41^\circ$$

$$X \sin 80.41 = 30$$

$$X = 30.42555317 \text{ mm}$$

$$\approx 30.43 \text{ mm}$$

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{9}{1}} = 3$$

$\therefore$ The response is:

$$x(t) = e^{-\frac{1}{6}(3)t} (30.43) \sin(3\sqrt{1-(\frac{1}{6})^2}t + 80.41^\circ)$$

$$= 30.43 e^{-\frac{1}{2}t} \sin\left(\frac{\sqrt{35}}{2}t + 80.41^\circ\right)$$

$$\approx 30.43 e^{-\frac{1}{2}t} \sin(2.96t + 80.41^\circ) \text{ mm}$$

$$\approx 30.43 e^{-\frac{1}{2}t} \sin(2.96t + 1.4) \text{ mm}$$